



The 8th Annual

Parallel Applications Workshop Alternatives to MPI+X

Sunday, November 16, 2025
9:00 am - 5:30 pm (CST)

Hybrid Event

Held in conjunction with SC25
The International Conference for High Performance Computing, Networking, Storage, and Analysis





Parallel Applications Workshop
Alternatives to MPI+X

November 16th, 2025
Sunday



Organization

- Workshop Chair
 - Karla V. Morris Wright - Sandia National Laboratories
- Organizing Committee
 - Engin Kayraklioglu - Hewlett Packard Enterprise
 - Kenjiro Taura - University of Tokyo
- Program Committee Chairs
 - Daniele Lezzi - Barcelona Supercomputing Center
 - Katherine Rasmussen - Lawrence Berkeley National Laboratory
- Artifact Evaluation Committee Chairs
 - Elliott Slaughter - SLAC National Accelerator Laboratory
 - Irene Moulitsas - Cranfield University
- Advisory Committee
 - Bradford L. Chamberlain - Hewlett Packard Enterprise
 - Damian W. I. Rouson - Lawrence Berkeley National Laboratory
 - Bill Long - Hewlett Packard Enterprise/Cray (Retired)



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Program Committee

- **Chair:** Daniele Lezzi - Barcelona Supercomputing Center
- **Chair:** Katherine Rasmussen - Lawrence Berkeley National Laboratory
- Marjan Asgari - National Resources Canada
- Dan Bonachea - Lawrence Berkeley National Laboratory
- Jan Ciesko - Sandia National Laboratories
- Irina Demeshko - NVIDIA
- Nils Deppe - Cornell University
- Nelson Dias - Federal University of Paraná
- Michael P. Ferguson - Hewlett Packard Enterprise
- Magne Haverlaen - University of Bergen
- Engin Kayraklioglu - Hewlett Packard Enterprise
- Bill Long - Hewlett Packard Enterprise/Cray (Retired)
- Francesc Lordan - Barcelona Supercomputing Center
- Nouredine Melab - University of Lille
- Esteban Meneses Rojas - National High Technology Center
- Henry Monge Camacho - Oak Ridge National Laboratory
- Karla V. Morris Wright - Sandia National Laboratories
- Irene Moulitsas - Cranfield University
- Julian Samaroo - Massachusetts Institute of Technology
- Michael Schlottke-Lakemper - University of Augsburg
- Elliott Slaughter - SLAC National Accelerator Laboratory
- Gabriel Tanase - Amazon Web Services
- Kenjiro Taura - University of Tokyo
- Miwako Tsuji - RIKEN Advanced Institute for Computational Science



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Artifact Evaluation Committee

- **Chair:** Irene Moulitsas - Cranfield University
- **Chair:** Elliott Slaughter - SLAC National Accelerator Laboratory
- Oliver Alvarado Rodriguez - Hewlett Packard Enterprise
- Keylor Arley - Universidad Costa Rica
- Yakup Budanaz - Technical University of Munich
- Fabio Durastante - University of Pisa
- Guillaume Helbecque - University of Lille
- Seema Mirchandaney - Stanford University
- Jonas Posner - University of Kassel
- Soren Rasmussen - National Center for Atmospheric Research
- Anjiang Wei - Stanford University



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Program

9:00	Welcome and Introduction
9:05	Session 1: Distinguished Talk Pavan Balaji - Meta
10:00	Break (30 minutes)
10:30	Session 2
12:30	Lunch (90 minutes)
2:00	Session 3
3:00	Break (30 minutes)
3:30	Session 4
4:30	Panel Discussion
5:30	PAW-ATM Concludes



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Session 1

9:00 - 10:00

9:00

Welcome and Introduction

9:05

Distinguished Talk

Pavan Balaji
Meta



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Session 2

10:30 - 12:30

10:30

Invited Talk

Alternative parallel programming models: past , present and is there a future?

Laxmikant Kale, University of Illinois at Urbana-Champaign

11:10

KVMSR+UDWeave: Extreme-Scaling with Fine-grained Parallelism on the UpDown Graph Supercomputer

Alexander Fell, Yuqing Wang, Tianshuo Su, Marziyeh Nourian, Wenyi Wang, Jose M. Monsalve-Diaz, Andronicus Rajasukumar, Jiya Su, Ruiqi Xu, Rajat Khandelwal, Tianchi Zhang, David F. Gleich, Yanjing Li, Hank Hoffmann, and Andrew A. Chien

11:30

Comparing Distributed-Memory Programming Frameworks with Radix Sort

Matt Drozt, Michael P. Ferguson, Ryan D. Friesse, and Shreyas Khandekar

11:50

Slicing Is All You Need: Towards A Universal One-Sided Algorithm for Distributed Matrix Multiplication

Benjamin Brock, and Renato Golin

12:10

DiOMP-Offloading: Toward Portable Distributed Heterogeneous OpenMP

Baodi Shan, Mauricio Araya-Polo, and Barbara Chapman



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Session 3

2:00 - 3:00

2:00

Invited Talk

***From Parallel Clusters to Hyper-distributed applications:
Programming swarms with COLMENA***

Francesc Lordan, Barcelona Supercomputing Center

2:20

Weak Scaling of NVSHMEM Applied To Hashed Distributed Structured Data

Andrew Davis, Hans Johansen, Xinfeng Gao, and Stephen Guzik

2:40

Redesigning GROMACS Halo Exchange: Improving Strong Scaling with GPU-initiated NVSHMEM

Mahesh Doijade, Andrey Alekseenko, Ania Brown, Alan Gray, and Szilárd Páll



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Session 4

3:30 - 5:30

3:30

Enhancing HPX with FleCSI: Automatic Detection of Implicit Task Dependencies

Davis Herring, Maxim Moraru, Scott Pakin, Julien Loiseau, Richard Berger, Philipp V. F. Edelmann, and Ben Bergen

3:50

Stackless vs. Stackful Coroutines: A Comparative Study for RDMA-based Asynchronous Many-Task (AMT) Runtimes

Mia Reitz and Jonas Posner

4:10

KDRSolvers: Scalable, Flexible, Task-Oriented Krylov Solvers

David K Zhang, Rohan Yadav, Alex Aiken, Fredrik Kjolstad, and Sean Treichler

4:30

Panel: Alternative programming models for applications at scale

Panel Chair: Anshu Dubey – Argonne National Laboratory

Ryan Coffee – SLAC National Accelerator Laboratory

Zhihao Jia – Carnegie Mellon University

Raye Kimmere - Princeton University

Peter Mendygral – Hewlett Packard Enterprise

Jeremy Wilke – NVIDIA

PAW-ATM Journey

